Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

## Phase 1 : Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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# 1. TEAM AND PROJECT OVERVIEW

**Team Name**

T010NG

**Selected Domain**

Domain 1: Aiversecity Future-Ready Workforce

**Selected SDG Track (1–17)**

SDG 4: Quality Education (target 4.4)

**Project / Solution Name**

Kotak

**Name (Leader)**

Hee Zi Jie

**Email & Phone (Leader)**

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**University Affiliation(s)**

Asia Pacific University of Technology & Innovation (APU), University of Reading and University of Sheffield

**Number of Team Members**

5

# 2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

## 2.1. The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.*

***[Maximum 250 words]***

AI is changing Malaysian jobs faster than training and skills lists can keep up. In Malaysia, 56% of professionals use AI tools at work, but only 26% have had formal training (Hays Asia Salary Guide 2025, via the Domain 1 brief). In October 2024, the government said about 600,000 workers in 10 sectors needed reskilling within three to five years (Bernama). TalentCorp projects about 620,000 workers, 18% of the workforce in the sectors it studied, will be significantly affected (Impact Study). A World Bank and ISIS Malaysia study rates 28% of the workforce highly exposed to AI (World Bank, 2025). Skills are also under-used: in Q2 2026, 35.2% of employed tertiary-educated workers held semi-skilled or low-skilled jobs (OpenDOSM; hiring-bar evidence).

Our user is a working adult in Malaysia, in any industry, who wonders what AI means for their job. They come with a worry and a few minutes, often on a phone. Some want to keep their job safe, some to move to a role they have in mind, and some to learn where they could go (users).

Today they get static lists. TalentCorp's MyMahir maps roles and skills, but shows no score, time estimate, scenario or SDG tag, and no Malaysian tool we checked joins role data to a scored gap and a timed pathway (prior art). Without dated evidence, a worker cannot tell which skills to learn first, and may spend scarce hours on tasks AI is taking over.

# 2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

## 2.2. The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.*

***[Maximum 300 words]***

Kotak, Malay for "box", is a web app. Kopi, a cat counsellor, asks 12 can-do questions, rated 1 to 4, about the user's job or a role they want. It draws a Future Skills Radar of the job's 6 core skills, ranks matched roles and builds a dated roadmap in about 3 minutes.

The six must-haves:

1. **Skills profile intake**: situation, study, a job title in any language, a goal (stay, move, explore or not sure) and the 12 questions.
2. **AI-trend signal layer**: dated Anthropic Economic Index snapshots for Malaysia (April and May 2026) and DOSM Job Market Insights, refreshed weekly.
3. **Future-role matching**: ranked roles, each with a fit score, its maths and named gaps.
4. **Reskilling pathway**: ordered steps, each with hours and one free resource, and a finish date set by the user's weekly hours.
5. **SDG mapping**: skills and jobs carry SDG targets, and the roadmap highlights steps by goal.
6. **Handling uncertainty**: targets are labelled estimates, Split shows slow and fast AI-uptake futures, and Unknown marks thin data.

# 2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

## 2.2. The Proposed AI Solution (continued)

Fixed rules, not AI, set every verdict, score, ranking and roadmap order. AI does four jobs:

- Before the demo, a RunPod GPU batch groups each O*NET job's own tasks into 6 skills and drafts the questions. A validator rejects any task outside that job.
- Live, Claude Haiku 5.5 explains each verdict with its dated sources and matches Malay and local job titles.
- Ask Kopi, a chat agent, answers through 8 read-only tools over the same rules, which other assistants reach through an MCP server.
- ElevenLabs voices Kopi, with mute and a text fallback.

From the Phase 1 workshops, we use GenAI APIs (Claude), Postgres (Neon), cloud hosting (Cloudflare Workers and Replit) and RunPod serverless GPUs (workshops; Workshop 1).

# 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

## 3.1. Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

Kotak is a TypeScript monorepo run with Bun. It has two apps, **apps/web** (the React app) and **apps/api** (the Hono API), and four packages: **shared** (the rules and Zod schemas), **sdk** (the check-up engine), **agent** (the AI tools) and **browser** (the JobStreet scraper's browser).

- **Frontend**: React and Vite, Tailwind CSS, Recharts for the radar and the roadmap charts, Lucide icons, and pixi-live2d-display for Kopi, built on Live2D's Hijiki sample model. It has phone (390 px) and desktop (1440 px) layouts, follows WCAG 2.2 AA and has a reduced-motion mode (design system).
- **Rule engine in the browser**: the app loads precomputed, dated JSON snapshots and computes levels, targets, verdicts, fit scores and roadmap order in the browser. Switching the snapshot from April to May 2026 recomputes them at once. A guest session in **localStorage** keeps answers, goals and ticked steps.
- **API**: Hono, on Cloudflare Workers for the main link and on Bun for Replit. One server serves the React build and the API. Its routes match job titles, write cited explanations, proxy text to speech, run Ask Kopi and serve an MCP endpoint. Zod validates every request and response. On Workers, the MCP and chat routes run in a Durable Object.
- **GenAI**: the Claude API with Claude Haiku 5.5. The API caches each answer by a SHA-256 hash of the request and pre-warms the cache for the demo roles. A check rejects any explanation that changes a verdict or lacks a dated citation, and falls back to a fixed template.

# 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

## 3.1. Conceptual Tech Stack (continued)

- **Agent layer**: **@kotak/agent** exposes 8 read-only tools over **@kotak/sdk**: find a role, get the questions, suggest roles, run a check-up, what changed, national skill gaps, job ads in Malaysia and list sources. Ask Kopi calls them through Claude, and outside assistants call them through **/api/mcp**. Every number the agent quotes comes from a tool.
- **Voice**: ElevenLabs. Fixed lines are pre-generated with **eleven_v4**, and live lines use **eleven_v4_turbo**. The API caches audio by a hash of the text, and the key stays on the server (TTS notes).
- **Data pipeline**: Bun scripts ingest O*NET 30.1, filter the Economic Index to Malaysia, Singapore and global rows, extract DOSM Job Market Insights (Q3 2019 to Q2 2026), and map fields of study to jobs through the NCES CIP to SOC crosswalk. Every record carries its source, date and layer. A GitHub Actions workflow checks for new releases every Saturday and rebuilds the snapshots.
- **Database**: Neon Postgres 18 (region **aws-ap-southeast-1**). One table holds the anonymous national view: only a role code, 6 skill levels, a sample flag and a date, with totals only for groups of 10 or more; no screen sends check-ups yet, so it holds labelled sample rows. Other tables hold the scraper's JobStreet listings, the admins' reviews and matched skills, and the admin accounts.
- **GPU**: a RunPod serverless endpoint runs **Qwen/Qwen3.8-27B-FP8** on vLLM for the one-time batch that builds 6 skills and 18 can-do statements for each of 923 jobs.
- **Hosting and CI**: GitHub Actions type-checks, builds and deploys Kotak to Cloudflare Workers on every merge to **main**, or when a member clicks "Run workflow" (deploy notes). Replit publishes the same **main** branch from the team's own account. API keys and the database URL live in GitHub, Cloudflare and Replit secrets, never in code.

System block diagram: see Appendix A.

Sponsor platforms: yes, Kotak uses both.

- **Replit**: the team publishes Kotak from **main** on its own Replit account at the live demo link. The project link, its history and the published app are the proof of use.
- **RunPod**: a serverless GPU endpoint, paid with the team's RunPod credit, ran the batch that built every job's skills and questions: a 3-job smoke test, then 920 jobs in 783 seconds, with none failing. Screenshots of the endpoint, GPU type and runtime are the proof of use (RunPod notes; sponsor rules).

# 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

## 3.2. Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.*

***[Maximum 200 words]***

- **Empathise**: on 21 October, we named three kinds of worker: stay strong, move to a target role, or ask where to go (users). Research showed Malaysian graduates find work, often below their skill level (hiring-bar evidence).
- **Define**: one user and one moment: a worried worker with a few minutes, who leaves with a plan to start this week.
- **Ideate**: we gave five AI models the brief, as other teams would, and 17 of their 75 ideas served this same worker (field simulation). So Kotak's uniqueness comes from its mechanism: rated can-do questions, skills built from the job's own tasks, and verdicts set by visible rules.
- **Prototype and test**: six AI-played experts raised 43 attacks on the first Kotak (roast). We cut the planted-mistake test and moved the demo from a sample card to real data. We set the visual design in Figma before coding.
- **Deploy**: the live demo swaps the Economic Index from April to May 2026, and the radar, ranking and roadmap move. Cached Claude answers and a fixed-template fallback cover API overload. GitHub Actions deploys to Cloudflare Workers on every merge to **main**, so any member can ship.

# 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.*

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

# 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

| Metric Category | Your Response |
| --- | --- |
| **Target SDG Sub-Target(s)**<br>*Identify the exact UN SDG sub-target your project addresses*<br><br>*e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."* | Target 4.4, "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship". It is measured by indicator 4.4.1, "Proportion of youth and adults with information and communications technology (ICT) skills, by type of skill" (UN SDG API, goal 4).<br><br>The target, indicator and baseline come from the UN SDG API, accessed on 22 October 2026, following our impact notes. |
| **Current Problem Baseline**<br>*What is the starting condition of the problem, expressed as a number?*<br><br>*e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."* | In 2024, 39.8% of Malaysians aged 15 and over had above-basic digital skills in all 5 skill areas, and 19.6% reported writing a computer program in a programming language. Both figures are indicator 4.4.1, series **SE_ADT_ACTS**, from DOSM (UN SDG API, Malaysia). For AI skills, 56% of Malaysian professionals use AI tools at work, but only 26% have had formal training (Domain 1 brief). |

# 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

| Metric Category | Your Response |
| --- | --- |
| **Core Measurable KPI**<br>*The specific unit you will track*<br><br>*e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided* | The sourced-roadmap rate. It is the percentage of finished check-ups that reach a roadmap in under 3 minutes, where every verdict, fit score and milestone shows its source and date. A second count tracks uncited recommendations, which must stay at 0 (LLM credibility notes). |
| **Phase 2 Impact Target**<br>*Your realistic, percentage-based target by the final prototype presentation*<br><br>*e.g., "20% improvement in computational resource use."* | 100% of 20 seeded test profiles reach a sourced roadmap in under 3 minutes with 0 uncited recommendations (LLM credibility notes). On 100% of the hand-checked demo roles, the April-to-May 2026 snapshot swap changes at least one verdict (demo roles). |

# 5. ELIGIBILITY & READINESS CHECKLIST

*Tick all boxes that apply before submitting.*

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

# 6. IP & ORIGINALITY DECLARATION

## 6.1. Pre-Existing / Prior-Owned Work

*Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?*

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. Kotak's code, design and research are new work by the team during the hackathon.

# 6. IP & ORIGINALITY DECLARATION

## 6.2. Third-Party Code, Datasets, or Models

*Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?*

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name | Type (code/dataset/model) | License |
|---|---|---|
| O*NET 30.1 Database | dataset | CC BY 4.0, with O*NET's fixed credit line and a modification notice source |
| Anthropic Economic Index | dataset | Data CC BY, version not stated; code MIT source |
| DOSM Job Market Insights | dataset | DOSM data terms: reuse with "Data source: Department of Statistics Malaysia" source |
| OpenDOSM skill-related underemployment | dataset | CC BY 4.0; one figure cited source |
| TalentCorp MyCOL and Epoch AI | dataset | Cited as context only, not loaded source |
| Qwen3.8-27B-FP8, run on RunPod | model | Apache 2.0 |
| Claude Haiku 5.5, through the Claude API | model | Proprietary API service from Anthropic |
| ElevenLabs text to speech | model | Proprietary API; users keep the rights to the output source |

# 6. IP & ORIGINALITY DECLARATION

## 6.2. Third-Party Code, Datasets, or Models (continued)

| Name | Type (code/dataset/model) | License |
|---|---|---|
| Live2D Hijiki sample model and Cubism Core | model (2D art) | Live2D Free Material License Agreement; Cubism Core under Live2D's own licence |
| Lucide icons | code | ISC |

The Live2D licence says "the Customer may not Redistribute all or part of the Material to third party" (clause 4.1.1). The team keeps the model in the repository and credits Live2D, a risk it accepted on 22 October 2026 (mascot decision).

The code also uses these open-source frameworks and tools: Bun, TypeScript, React, Vite, Tailwind CSS, Recharts, pixi.js, pixi-live2d-display, Hono, Zod, the Anthropic TypeScript SDK, the MCP server SDK, the Neon serverless driver, Cloudflare's Wrangler, Playwright and Clearcote. Every licence is listed in the technical declaration and in NOTICE.md. Kotak runs on hosted services that add no code to the repository: Cloudflare Workers, Neon Postgres, GitHub Actions and Replit.

# 6. IP & ORIGINALITY DECLARATION

## 6.3. Employment-Related IP Conflicts

*Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?*

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable. No team member is bound by an employment, internship or consulting agreement that lets an employer claim work made during the hackathon.

# 7. DECLARATION

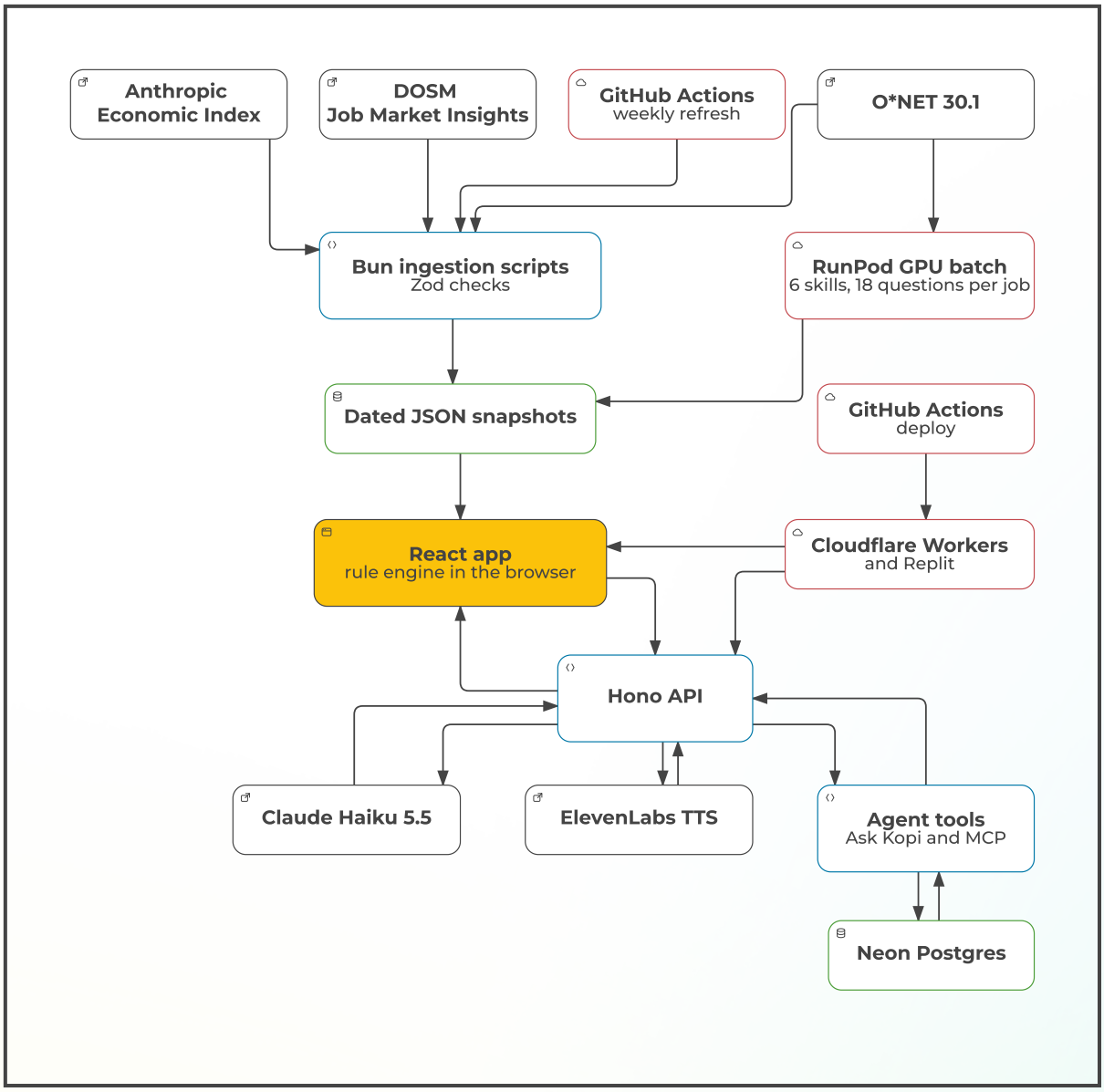
By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24 October 2026

# 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

## A. Appendix A: System Block Diagram




Public data, weekly refresh and RunPod feed dated snapshots for the browser rules. Deployment serves React and Hono. Hono connects Claude, ElevenLabs and agent tools; the tools connect Neon Postgres.

# 8. CHANGES FOR THE FINAL

## 8. Changes for the final

What changed after the preliminary round, and which feedback asked for each change.

| Change | Why | Feedback behind it |
|---|---|---|
| Every SDG chip names its target, gives a one-line reason and the UN text, and the roadmap totals the plan's targets | A bare "SDG 4" told the user nothing, and a badge that changes no choice is the guide's "not enough" | Finalist guide must-have 5 |
| 1,838 of 5,520 skill corners were judged against each target's UN text, then retagged or kept (docs/sdg-retag-review.md) | Some reasons shown were not true of the skill under them | Team review |
| The roadmap scales each step's hours by the user's situation, experience and qualification | Two people with the same answers got the same plan | Team review |
| The Where You Stand card shows four named boxes, dots that open on tap, and a table version | The dots crowded one box and needed the list below to decode | Team review |
| The questions page holds still: one speech box size, a pink chosen answer, no jump while a font loads | The answers moved between questions and on a cold first visit | Team review |

# 8. CHANGES FOR THE FINAL

## 8. Changes for the final (continued)

| Change | Why | Feedback behind it |
|---|---|---|
| Arrow keys move through the job, target-role and top-bar search suggestions | Each suggestion was its own Tab stop, so leaving took up to 13 presses | Team review |
| Kopi keeps Claude's cited summary when no verdict changed, not the fixed template | The guardrail rejected drafts the prompt allowed | Team review |
| A published cost: about 7 US cents per check-up, about US$1,900 to US$7,000 a month at 100,000 users (docs/cost-card.md) | "How will Kotak keep running?" had no figure behind it | Mentor point 1: explain the business model |
| Smaller fixes: sign in before the check-up, readable verdict chips on the yellow card, an (i) popover off the answers | Each was a break a user could see | Team review |

**Not changed, and why**

- **The employer team view.** Not built. The team presents it as the next step, never as a working feature.
- **Testing Kotak on 3 to 5 real people.** Not done. It needs people outside the team, not code.
- **"What suits the company you want to join."** Out of scope. It stays with pasting a job ad, which the team chose not to build (docs/decision-log.md).

# 8. CHANGES FOR THE FINAL

## 8. Changes in the final sprint, 31 October to 1 November

| Change | Why | Feedback behind it |
|---|---|---|
| A roadmap step counts as done only once the user proves it: an uploaded certificate that Claude judges pass, fail or unclear | A ticked box proved nothing | Mentor session, 31 October |
| Every job title, and the demo jobs in full, in Malay and Chinese | Picking Malay left a job's own text in English | Team review |
| SkillsFuture Singapore's name and level (1 to 6) beside each skill, and each job's Singapore equivalent; Claude Haiku 5.5 matched 2,030 Singapore roles to 304 O*NET jobs | O*NET is a US list, and the skills were drafted from US tasks | A SkillsFuture ambassador at the venue, 31 October |
| Verdict chips explain themselves on hover; Interpret asks Kopi what a gap is, why it matters and how to close it | First-time users didn't know what the chips meant | First-time users the team asked |
| Ask Kopi opens mid-screen, the plan bar fades before the steps, and the date picker opens under its field | The chat was cramped, the bar covered the steps and the picker broke the page | Team review |